

DSAgent

Autonomous Pipeline Report

Dataset: titanic.csv

Generated: May 19, 2026 at 11:00 PM

Session: 8af261af-2ab...

19

Steps Executed

6

Phases Completed

33.1s

Total Time

1. Dataset Overview

The dataset contains **891** rows and **12** columns, using **0.28 MB** of memory.

Type	Count	Columns
Numeric	7	PassengerId, Survived, Pclass, Age, SibSp, Parch, Fare
Categorical	5	Name, Sex, Ticket, Cabin, Embarked

Numeric Statistics

Column	Mean	Median	Std	Min	Max
PassengerId	446.00	446.00	257.35	1.00	891.00
Survived	0.38	0.00	0.49	0.00	1.00
Pclass	2.31	3.00	0.84	1.00	3.00
Age	29.70	28.00	14.53	0.42	80.00
SibSp	0.52	0.00	1.10	0.00	8.00
Parch	0.38	0.00	0.81	0.00	6.00
Fare	32.20	14.45	49.69	0.00	512.33

AI Analysis

Key Insights About the Dataset • Dataset Overview: The dataset contains 891 rows and 12 columns, with a mix of numeric and categorical features. It appears to be a classic Titanic dataset, likely from the 1912 sinking. • Target Variable: The Survived column is a binary variable (0 = died, 1 = survived), which is the likely target variable for a classification task. • Key Variables: • Pclass (Passenger Class): Shows a moderate negative correlation with Survived, suggesting that lower class (1st) passengers had a higher survival rate. • Age: Moderately negatively correlated with Survived, indicating younger passengers were more likely to survive. • Fare: Weakly positively correlated with Survived, suggesting that passengers who paid higher fares may have had better access to lifeboats. • Sex: While not in the correlation matrix, Sex (categorical) is a strong predictor of survival, especially for females. • Embarked (Port of Embarkation): Only 2 missing values, with most passengers boarding at Southampton (S). • Missing Data: 866 total missing values across 3 columns, with Cabin being the most problematic (77.1% missing), followed by Age (19.87%). Data Quality Issues Found • High Missing Values: • Cabin: 77.1% of values are missing. This column may not be useful for modeling due to lack of data. • Age: 19.87% missing. This is a key demographic variable, so imputation or removal of missing rows may be needed. • Embarked: Only 2 missing values. These can be imputed or removed with minimal impact. • High Cardinality: • Name has 891 unique values, making it impractical for modeling unless used for feature engineering (e.g., extracting titles). • Low Cardinality Categorical Columns: • Sex and Embarked are low cardinality and can be encoded easily. • Data Types: • PassengerId is numeric but not useful for modeling. • Ticket is an object type with high cardinality and likely not useful. Columns That Look Most Interesting/Useful • Survived: Target variable for classification. • Sex: Strong predictor of survival. • Pclass: Strong negative correlation with survival. • Age: Moderately correlated with survival. • Fare: Weakly correlated with survival. • Embarked: May provide some geographic context. • SibSp and Parch: Could indicate family relationships, which may influence survival (e.g., children or family members being prioritized). Likely Target Variable (for ML) • Target Variable: Survived (binary classification task). • Reasoning: The Survived column is the only binary variable in the dataset and is the most logical outcome to predict. It is also

the most commonly used target in the Titanic dataset for machine learning tasks.

Summary of Recommendations • Impute or drop missing values in Age and Cabin. • Encode categorical variables like Sex, Embarked, and Ticket (if needed). • Drop PassengerId and Name (unless used for feature engineering). • Focus on Pclass, Sex, Age, Fare, and SibSp for modeling. • Consider using Title extracted from Name as a new feature for better performance.

2. Data Quality Assessment

Found **3** columns with missing values out of 891 rows.

Column	Missing Count	Missing %
Age	177	19.87%
Cabin	687	77.1%
Embarked	2	0.22%

Potential Issues:

- High cardinality categorical columns: ['Name']

3. Data Cleaning Actions

Age has 19.87% missing

column: Age
strategy: median
action: Filled with median: 28.00
nulls_before: 177
nulls_after: 0
rows_after: 891

Embarked has 2 missing values

column: Embarked
strategy: mode
action: Filled with mode: S
nulls_before: 2
nulls_after: 0
rows_after: 891

Cabin has 77.1% missing and is not useful for modeling

Fare has high variability and potential outliers

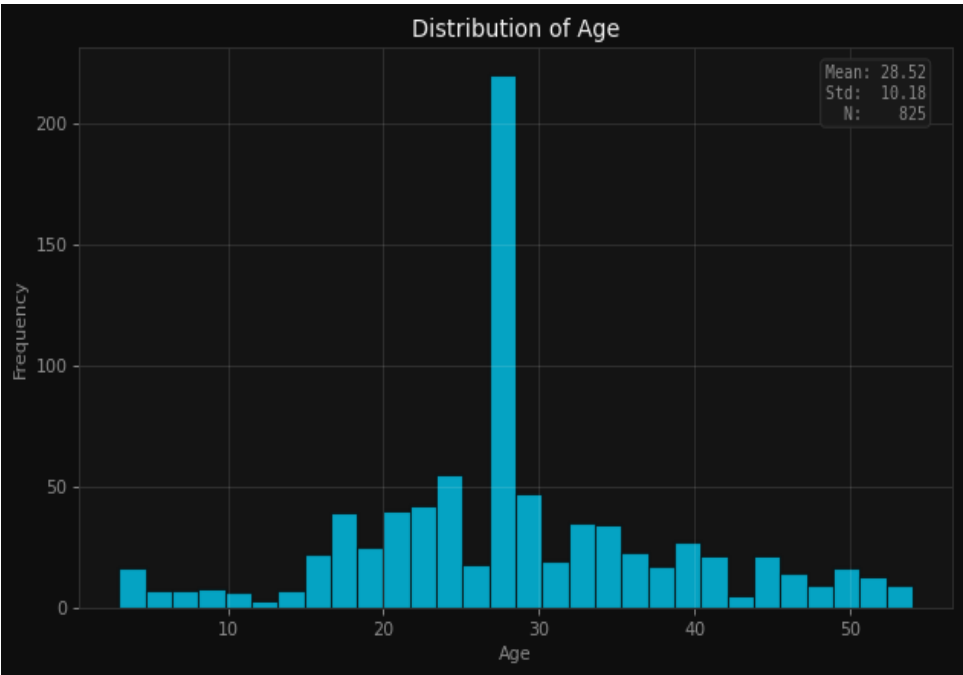
column: Fare
method: iqr
threshold: 1.5
outlier_count: 116
outlier_percentage: 13.02
total_rows: 891

Age has potential outliers

column: Age
method: iqr
threshold: 1.5
rows_removed: 66
rows_remaining: 825
removal_percentage: 7.41
Applied 5 cleaning operations based on data quality analysis.

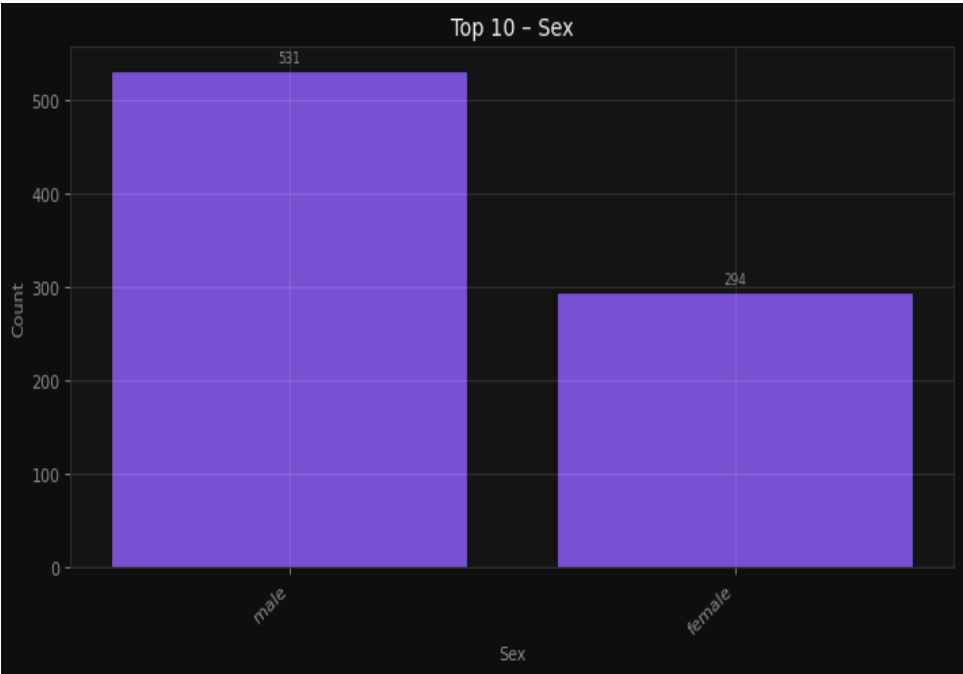
4. Visualizations & Insights

Understand the age distribution of passengers



Understand the age distribution of passengers

Compare the count of male and female passengers



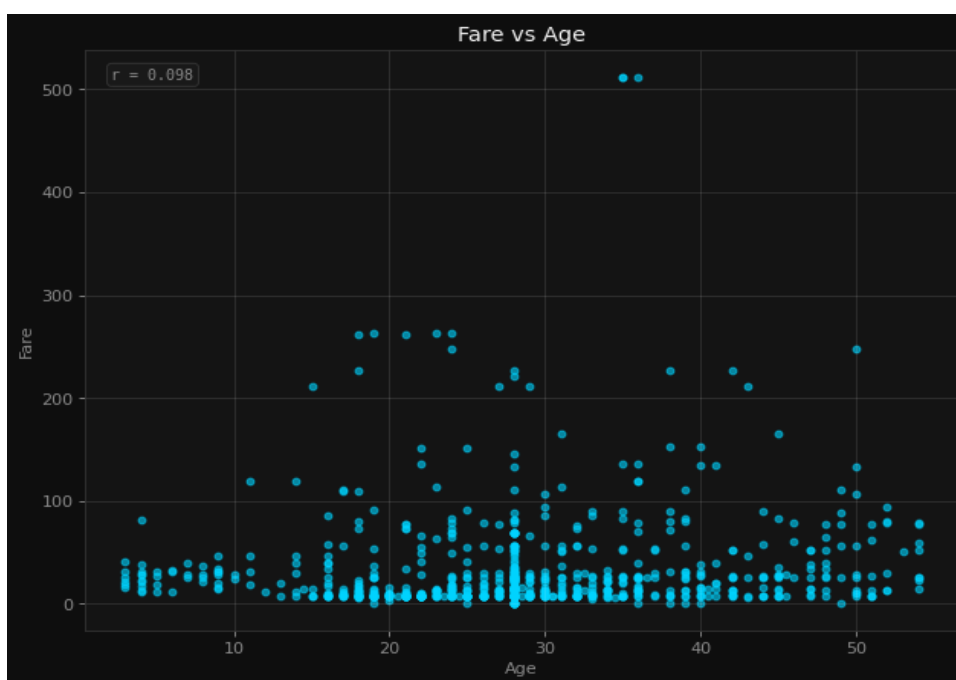
Compare the count of male and female passengers

Identify outliers in fare payments



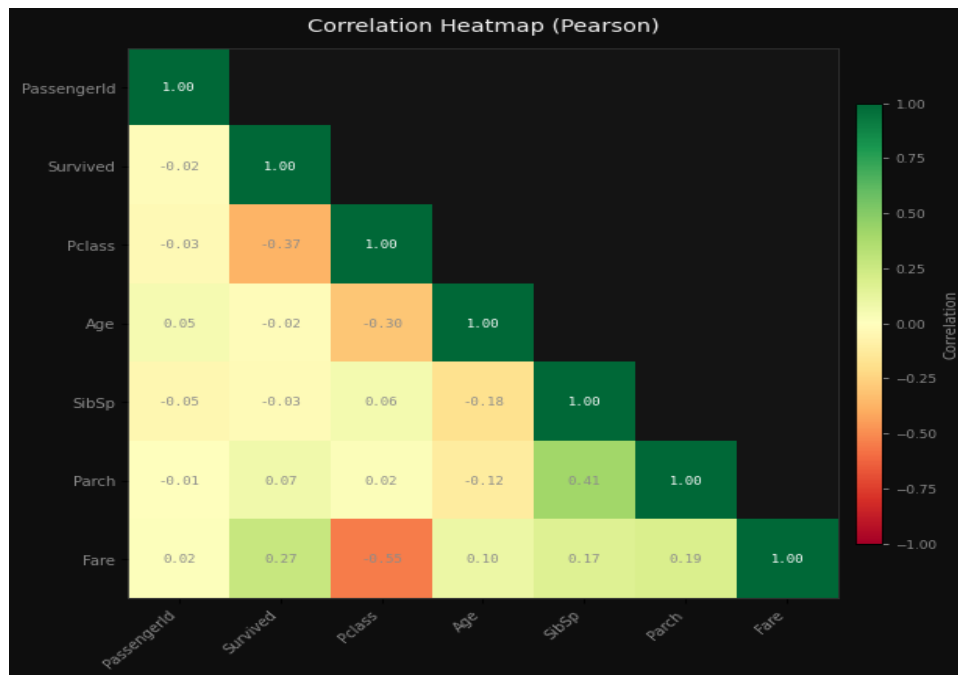
Identify outliers in fare payments

Explore the relationship between age and fare



Explore the relationship between age and fare

Identify correlations between numeric features



Identify correlations between numeric features

AI Interpretation

Histogram (Age Distribution): Likely shows a right-skewed distribution with a peak around the mid-20s to 30s, reflecting the typical age range of passengers on the Titanic. It may also highlight the presence of missing values, as the dataset has many missing ages. Bar Chart (Male vs Female): Likely reveals a significant majority of male passengers, as historical context suggests more men were on board, and the dataset has a higher number of males than females. Box Plot (Fare Outliers): Likely shows a few high fare outliers, given the presence of missing values and the potential for extreme values in the fare distribution, especially among first-class passengers. Scatter Plot (Age vs Fare): Likely reveals a weak or no clear correlation, with some clustering of higher fares among older passengers or those in higher classes, but not a strong linear relationship. Correlation Heatmap: Likely shows low or no correlation between most numeric features, with the strongest correlation possibly between Pclass and Fare (as lower classes tend to pay less), and potentially a weak correlation between Age and Fare.

5. Feature Engineering

Encode categorical variables for ML modeling

```
encoding: One-Hot (get_dummies)
drop_first: True
columns_before: 12
columns_after: 13
new_columns_added: 1
rows: 825
```

Normalize numeric features for better model performance

```
scaler: StandardScaler (Z-score)
rows: 825
note: Dataset updated in session. Features now have  $\mu \approx 0$ ,  $\sigma \approx 1$ .
Applied 2 feature engineering steps.
```


6. Model Training & Comparison

Model Selection Rationale

LLM error: 402

7. Model Evaluation

Model Evaluation Metrics

Model evaluated with full metrics and diagnostic plots.

8. Conclusions & Recommendations

LLM error: 402